Ryerson University

Department of Electrical, Computer & Biomedical Engineering
Faculty of Engineering & Architectural Science

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Instructor	Arman Aghaee

Lab	04
Lab Title	

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Simulation Exercise 1

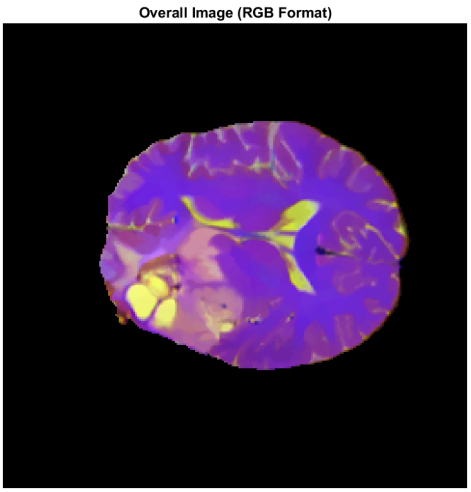


Figure 1.1. Overall Image of Brain (RGB Format)

1.2 - Estimate the number of regions by examining the overall image. Explain your answer in 2-4 sentences.

By visually inspecting the RGB image, the number of distinct regions can be estimated based on clearly separable colour/intensity groupings. In this case, the image appears to contain approximately 3-4 dominant regions, corresponding to major colour clusters (e.g., background, mid-tone tissue/area, and brighter structures).

This makes sense because segmentation aims to partition an image into meaningful regions based on intensity or colour similarity . If the colours are well separated visually, they will likely form distinct clusters in feature space.

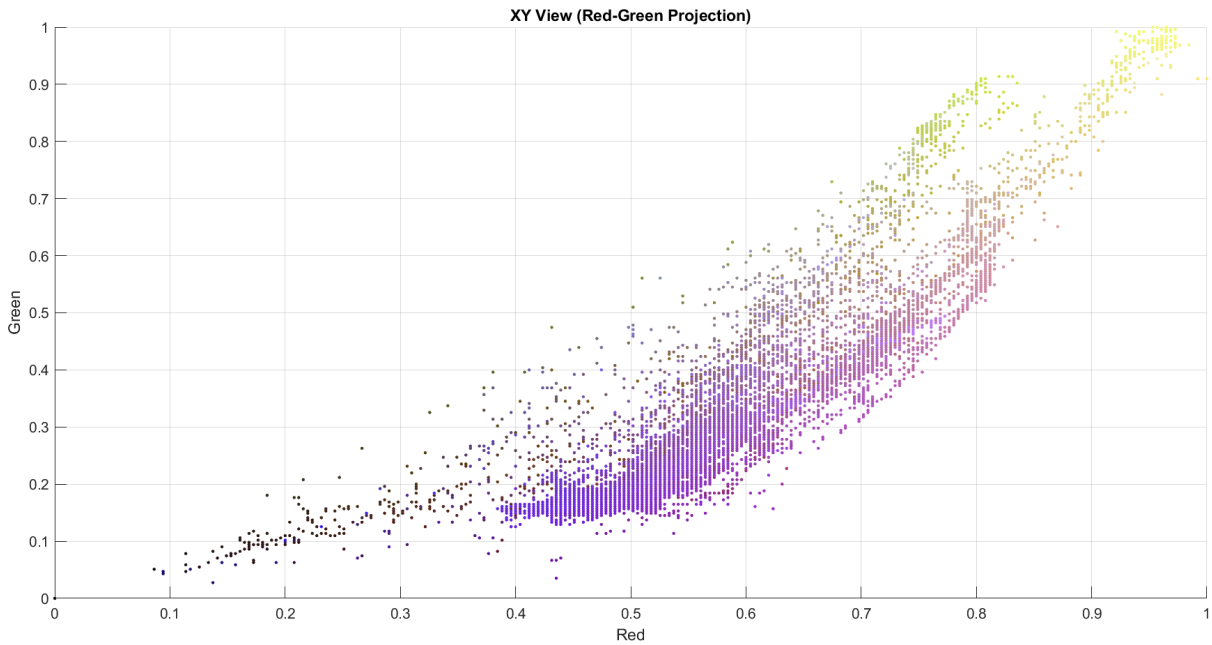


Figure 1.3.1. XY (RG) View of 3D Space

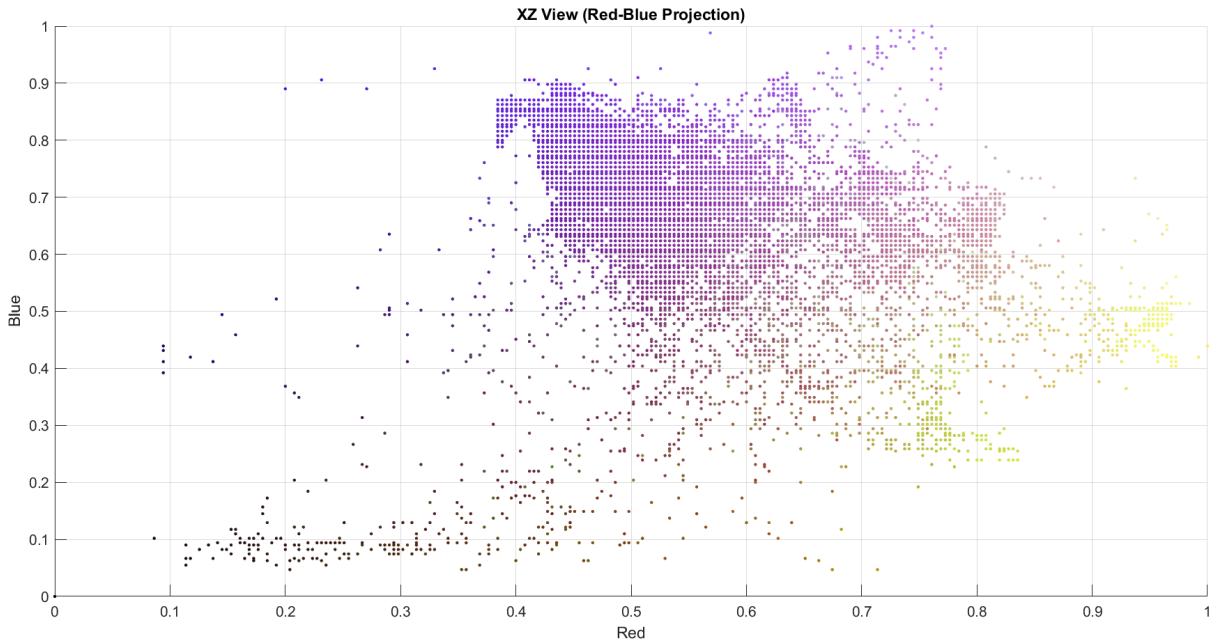


Figure 1.3.2. XZ (RB) View of 3D Space.

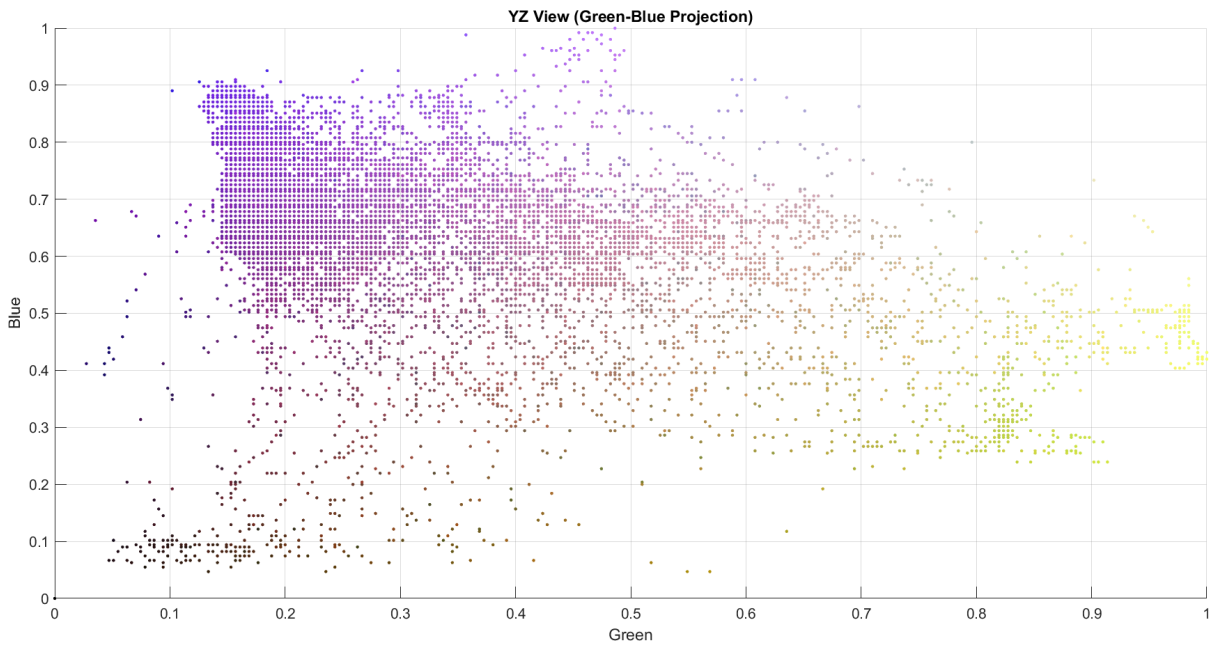


Figure 1.3.3. YZ (GB) View of 3D Space

1.4 - By examining the data in this space, check if the number of regions (clusters) determined in the previous step is appropriate.

When the image is plotted in 3D RGB space, each pixel becomes a point in (R,G,B). If the number of regions chosen earlier is correct, we should see distinct clusters (dense point groupings). From examining this space, the clusters appear reasonably separated but with some overlap, suggesting that the chosen number of regions is mostly appropriate, though slight overlap indicates natural variation or noise in pixel values. This aligns with

segmentation theory, where real images rarely produce perfectly separable clusters due to noise and intensity variation

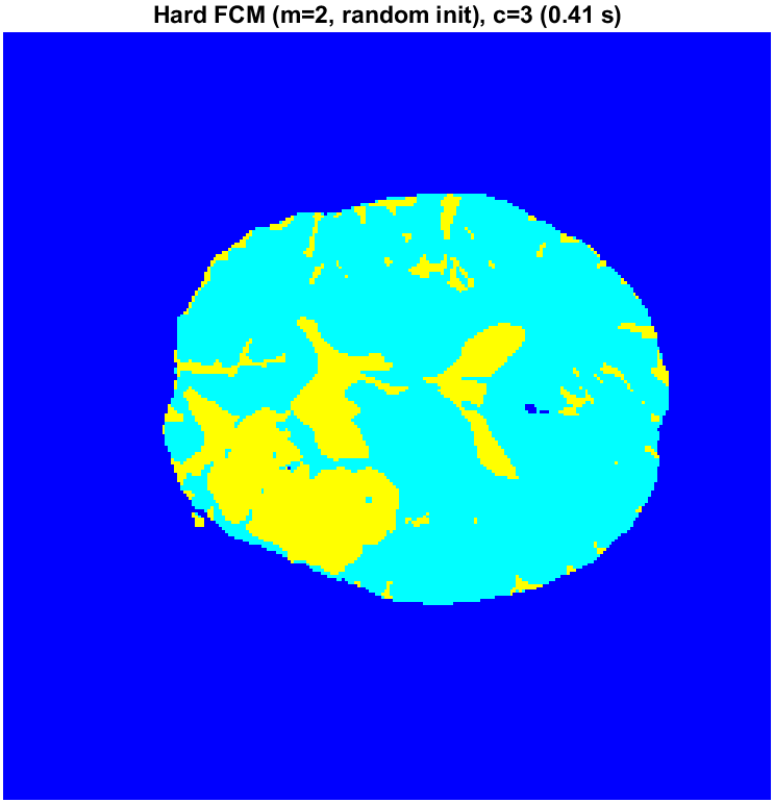


Figure 1.5. Hard FCM of the Brain

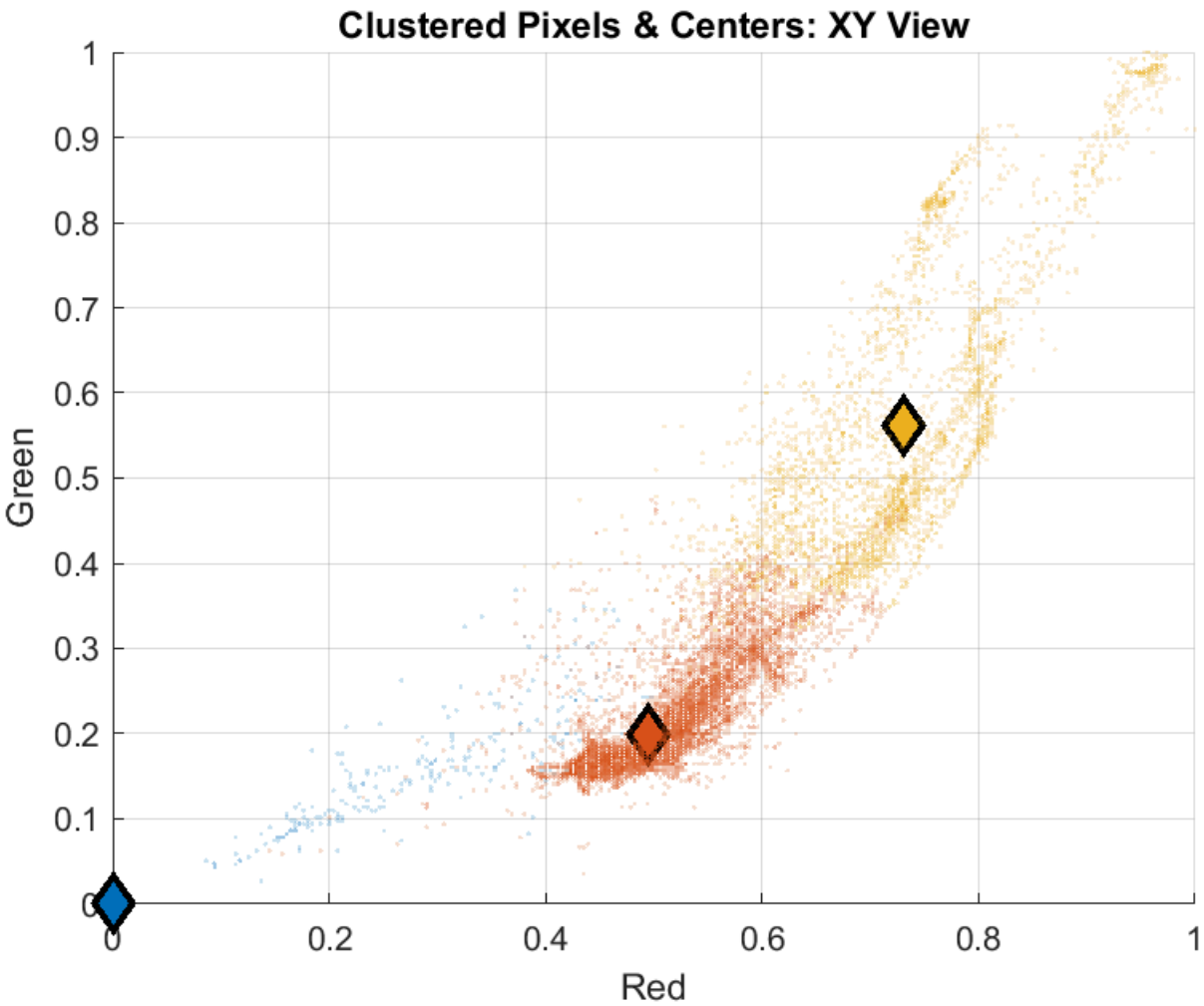


Figure 1.6.1 Clustered pixels and centers, XY View

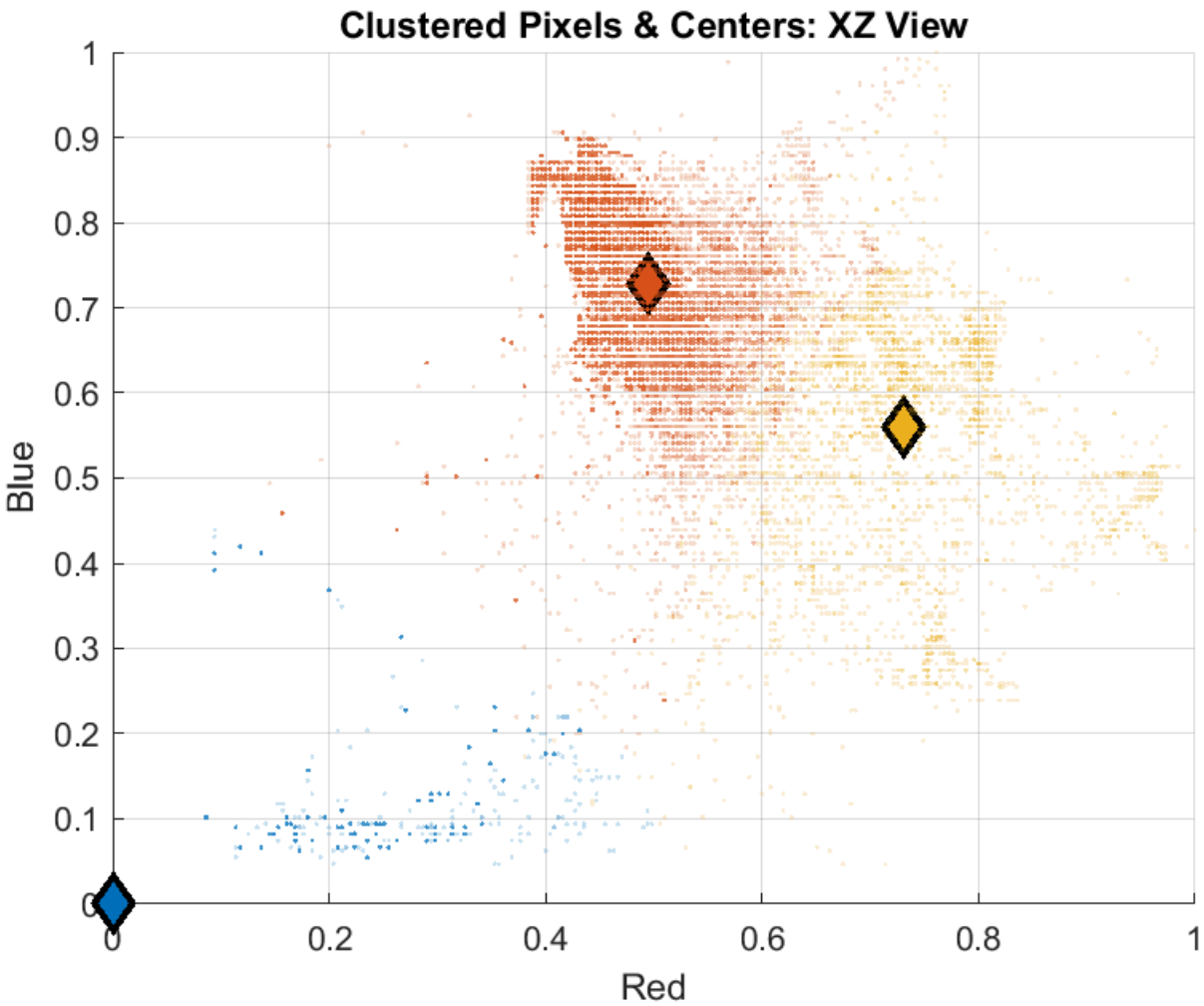


Figure 1.6.2 Clustered pixels and centers, XZ View

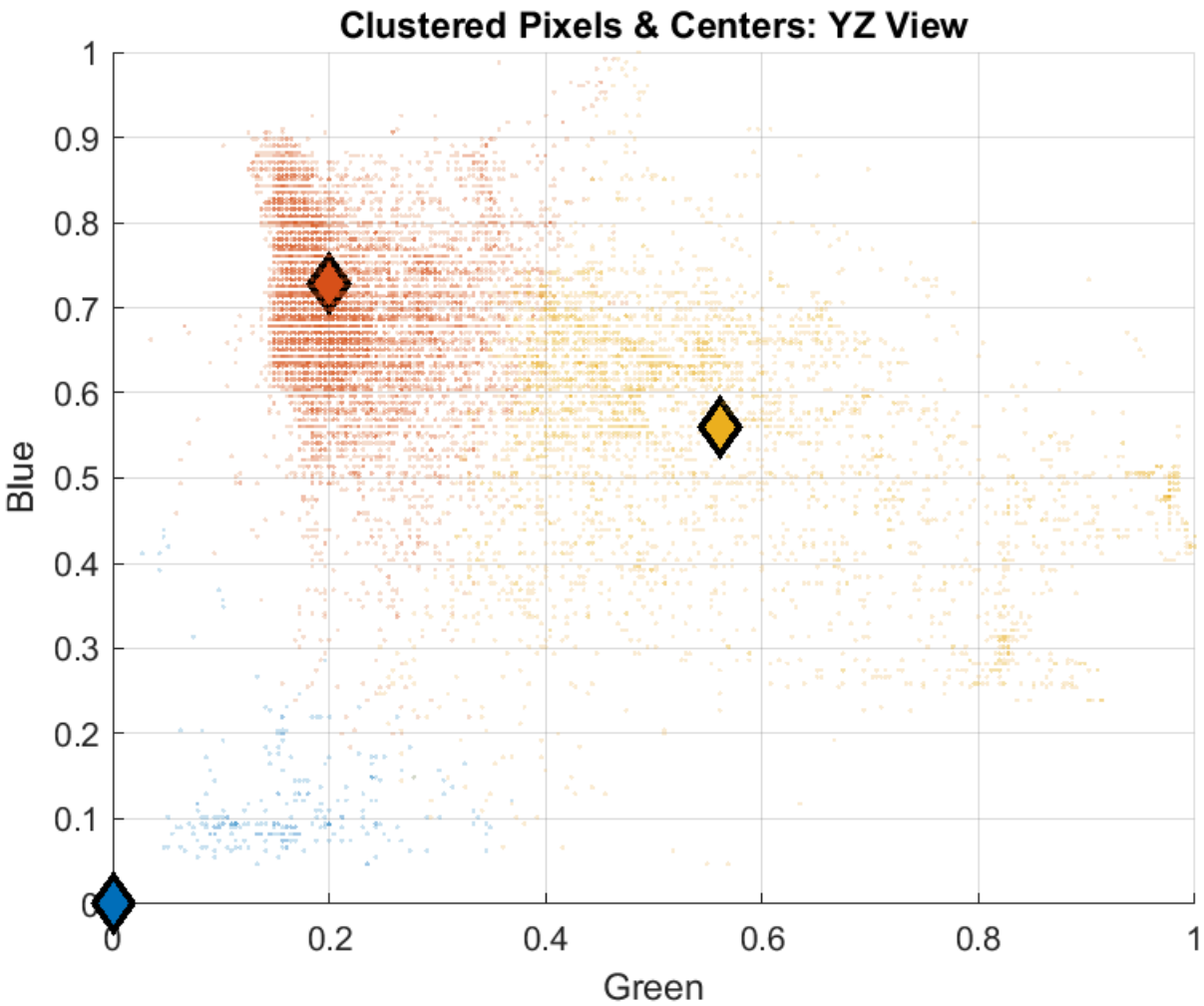


Figure 1.6.3 Clustered pixels and centers, YZ View

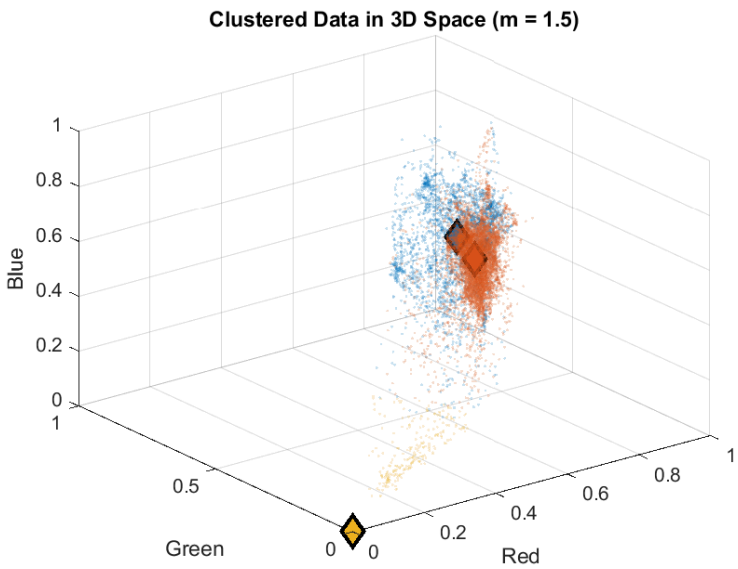


Figure 1.7.1 Clustered pixels and Centers, m=1.5

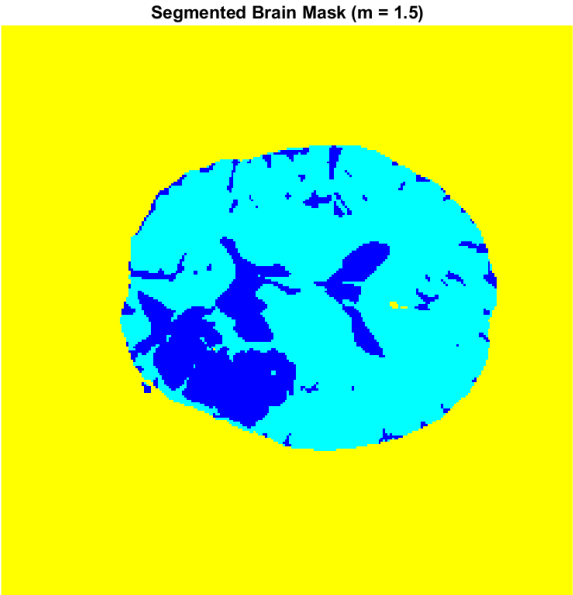


Figure 1.7.2 segmented Brain Mask, m=1.5

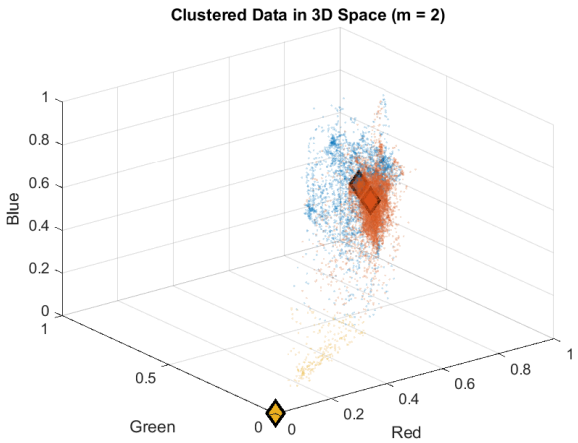


Figure 1.7.3 Clustered pixels and Centers, m=2

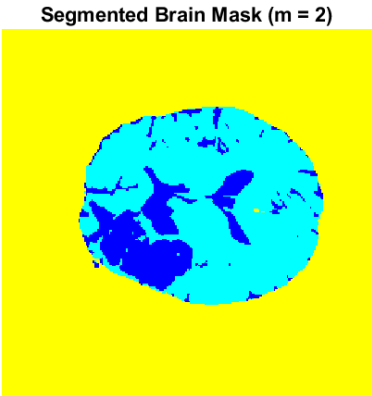


Figure 1.7.4 segmented Brain Mask, m=2

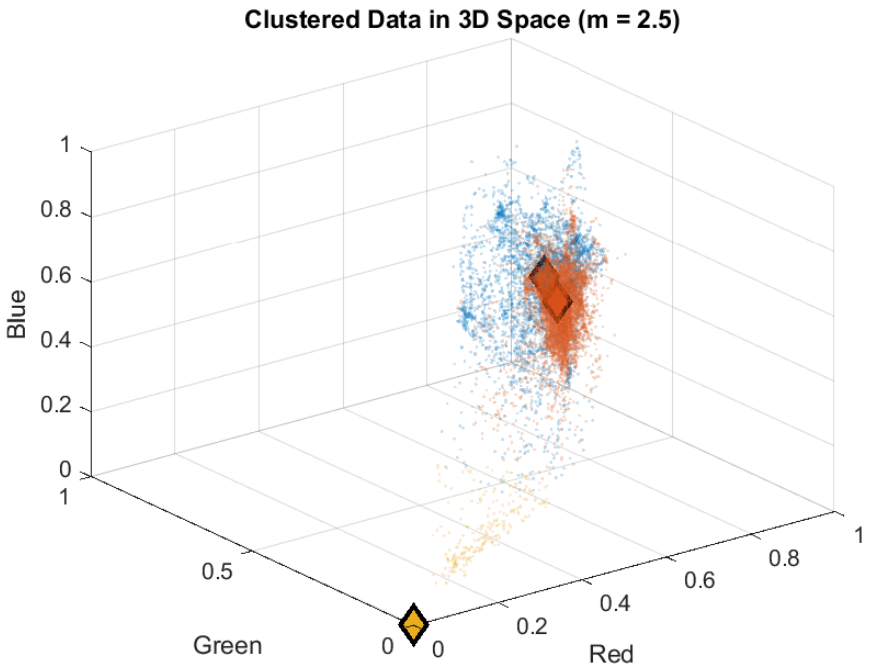


Figure 1.7.5 Clustered pixels and Centers, m=2.5

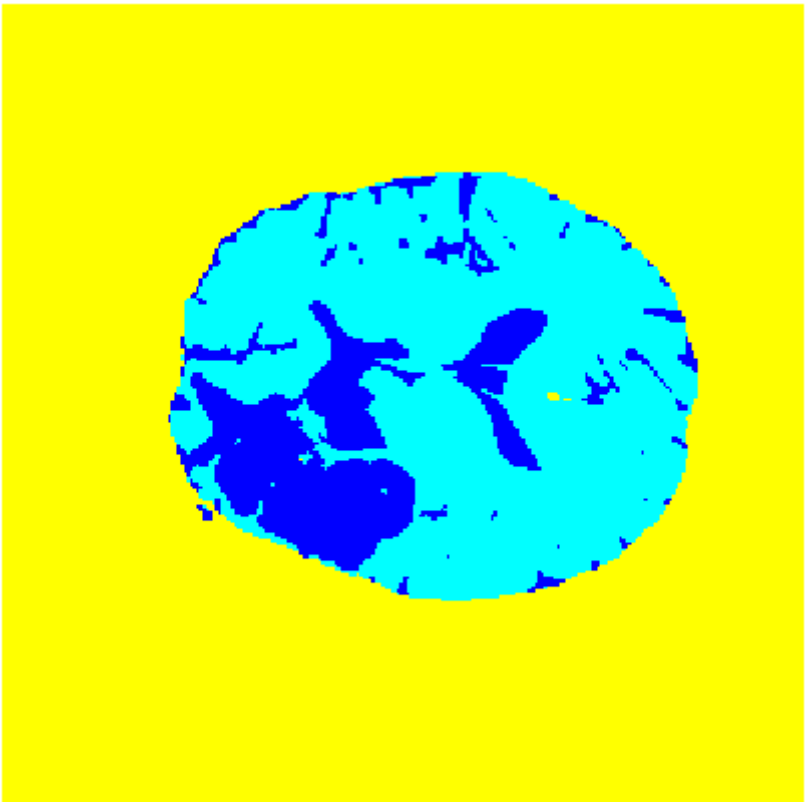
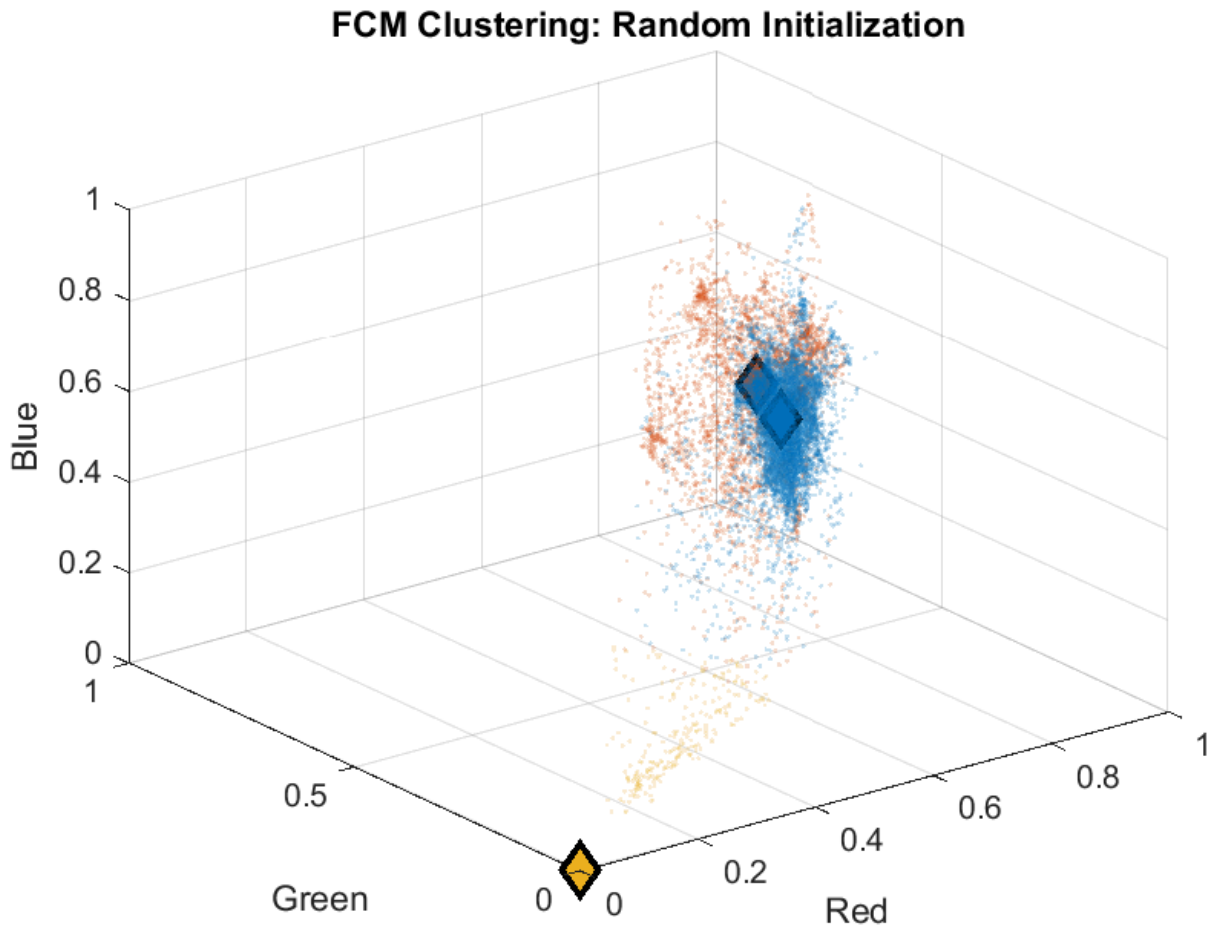


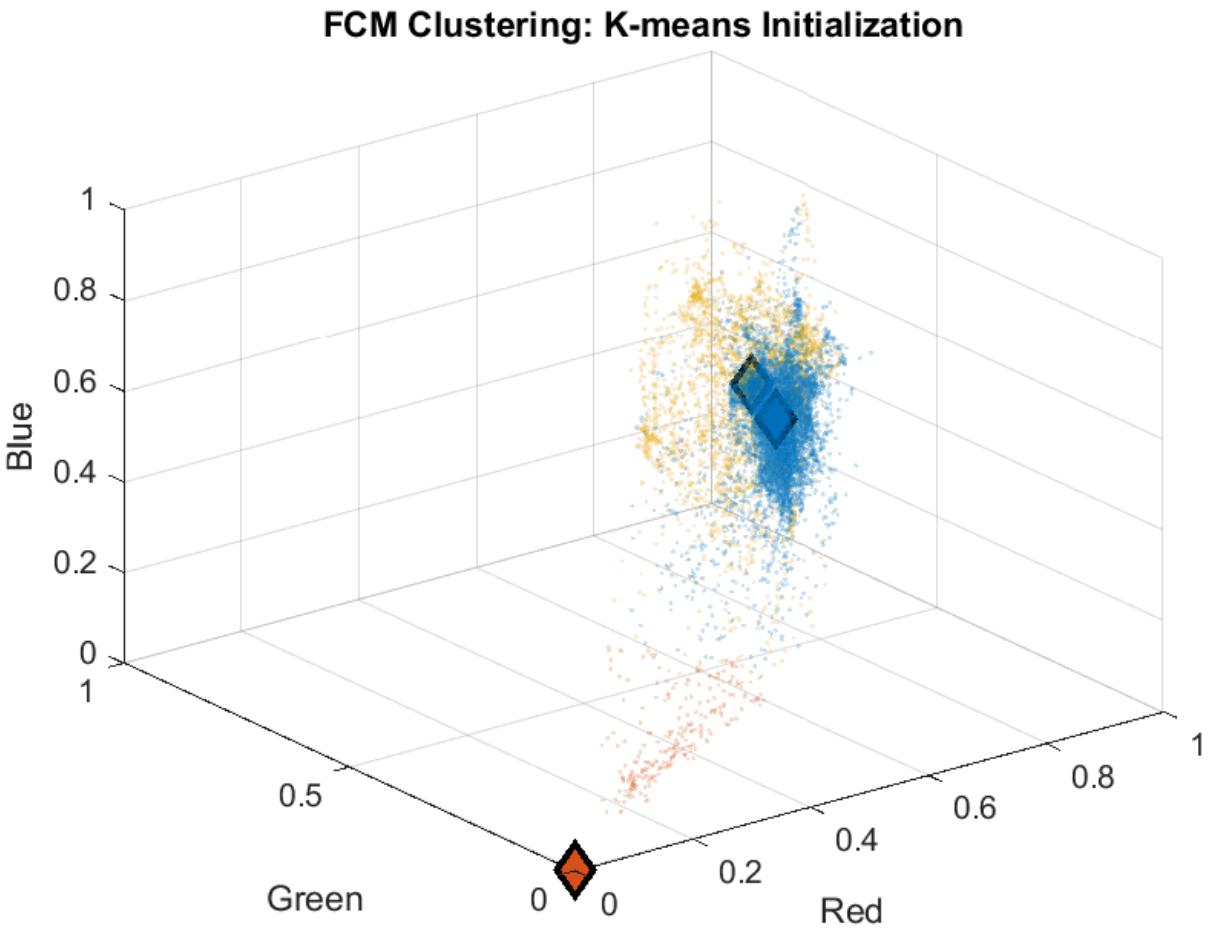
Figure 1.7.6 segmented Brain Mask, m=2.5

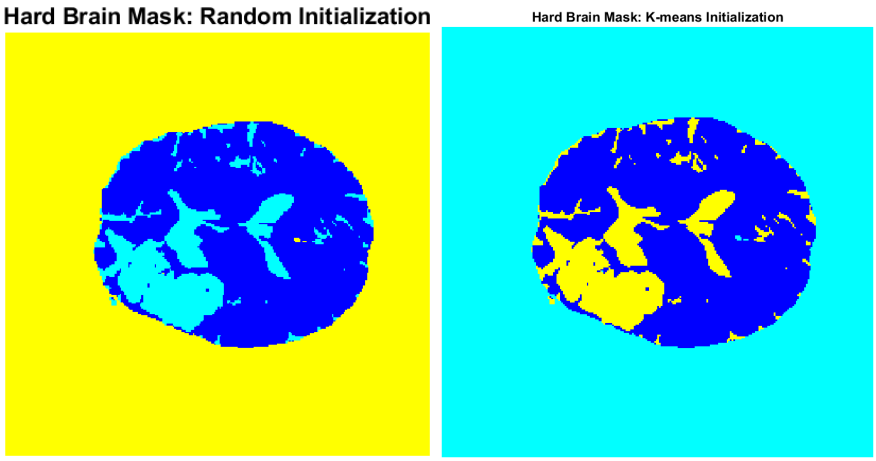
1.7 - Explain what the exponent for the fuzzy partition matrix does, which one produced the best clustering, and why. Explain the region of interest.

The exponent m controls how "fuzzy" the clustering is, meaning how much overlap there is between clusters. When m is lower (like 1.5), the clustering becomes more strict and starts to

behave like hard clustering, which can make boundaries noisy or too sharp. When m is higher (like 2.5), the clusters become too blended and lose clear separation. In my results, m = 2 gave the best balance, since it kept regions distinct while still handling gradual transitions. This was especially noticeable in boundary areas (ROI), where m = 2 preserved structure without over-smoothing or introducing noise.





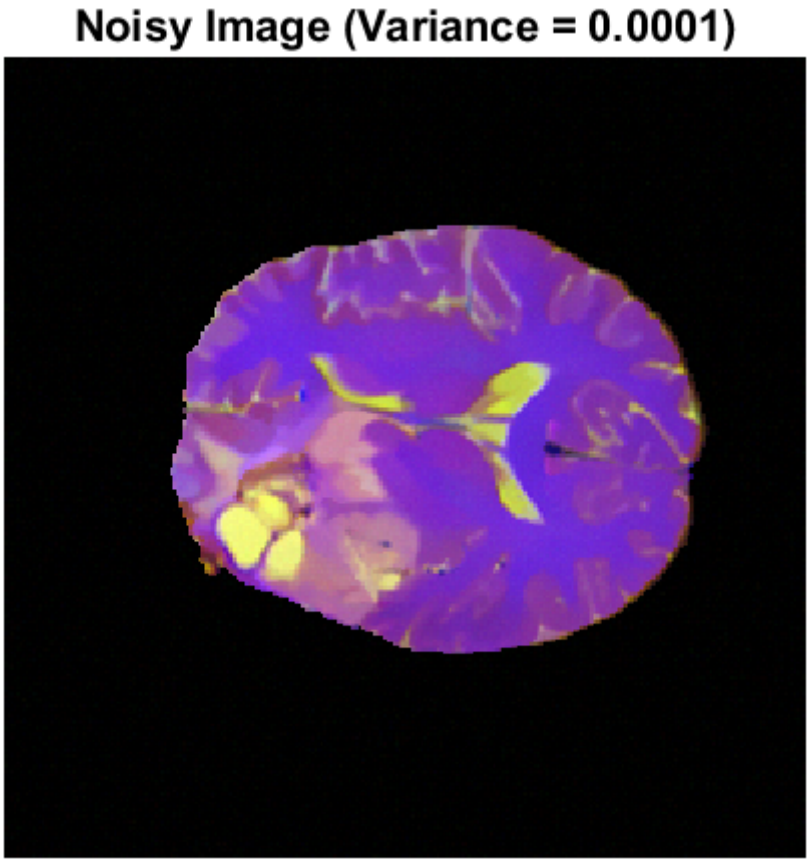


1.8 - Discuss differences between K-means clustering and randomized clustering

K-means clustering is more stable because it finds cluster centres in a structured way, while randomized clustering depends heavily on initial conditions and can give inconsistent results. In my case, K-means produced cleaner and more well-defined regions, while the randomized approach sometimes misclassified pixels or created uneven clusters. This difference was most

obvious in mixed or boundary regions (ROI), where K-means handled separation better and gave more consistent segmentation overall.

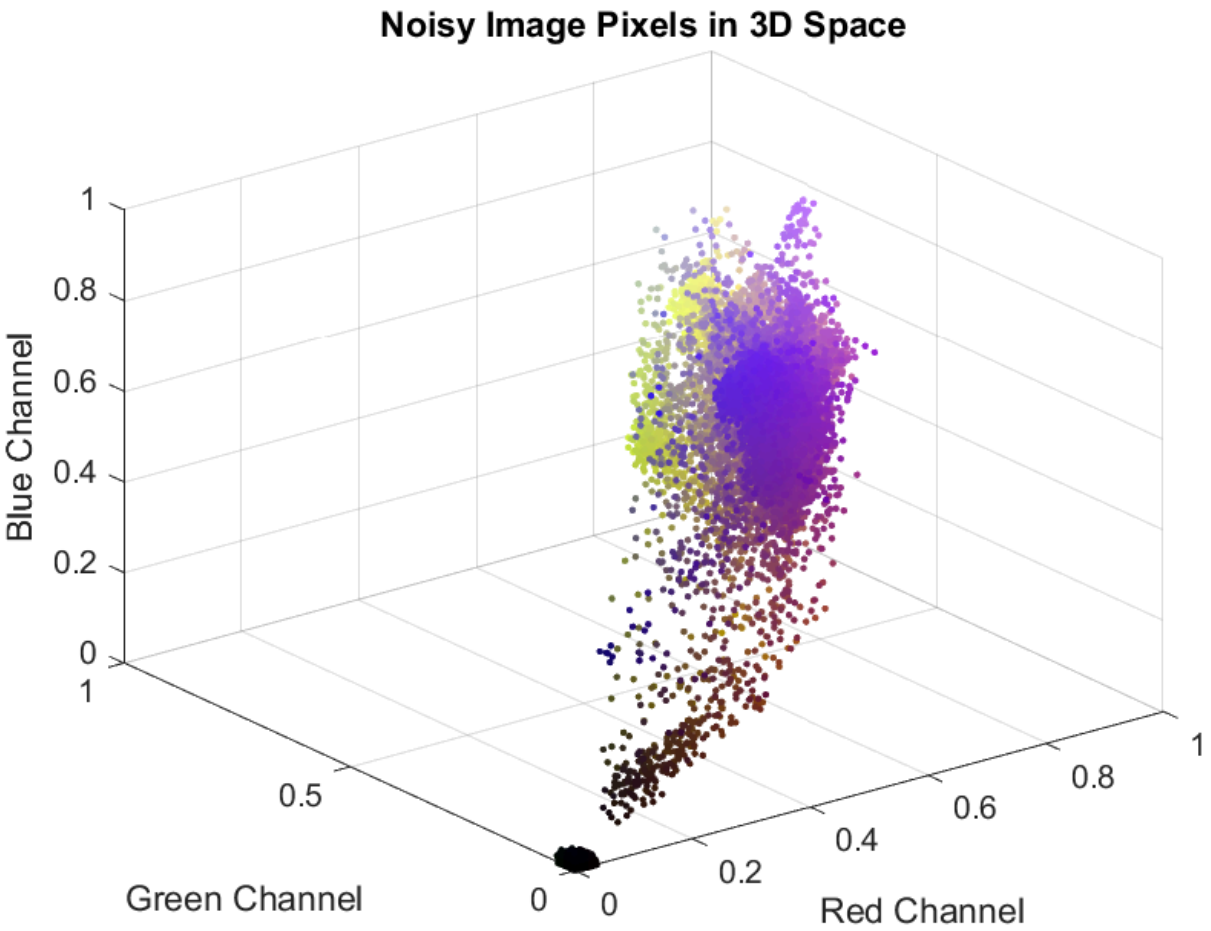
Simulation Exercise 2

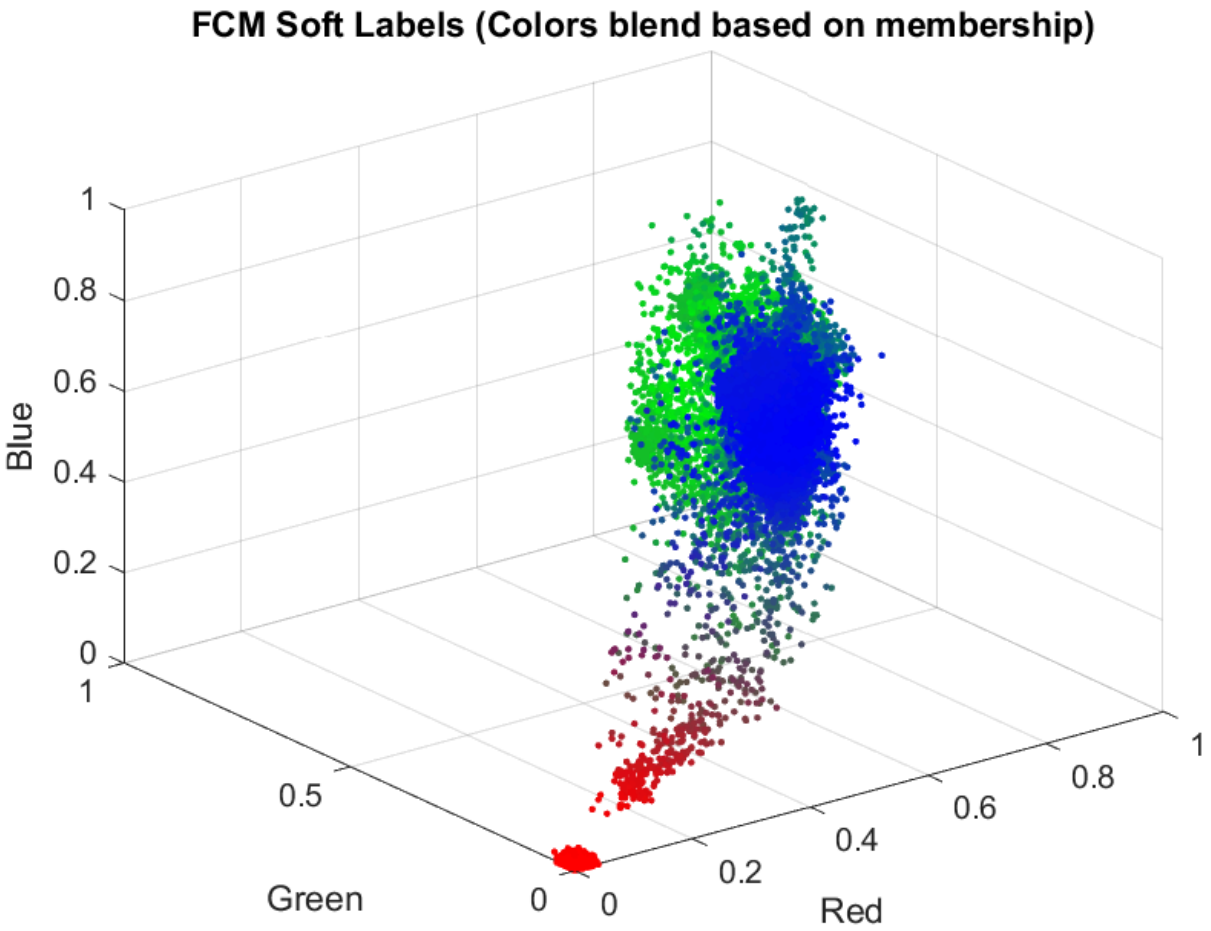


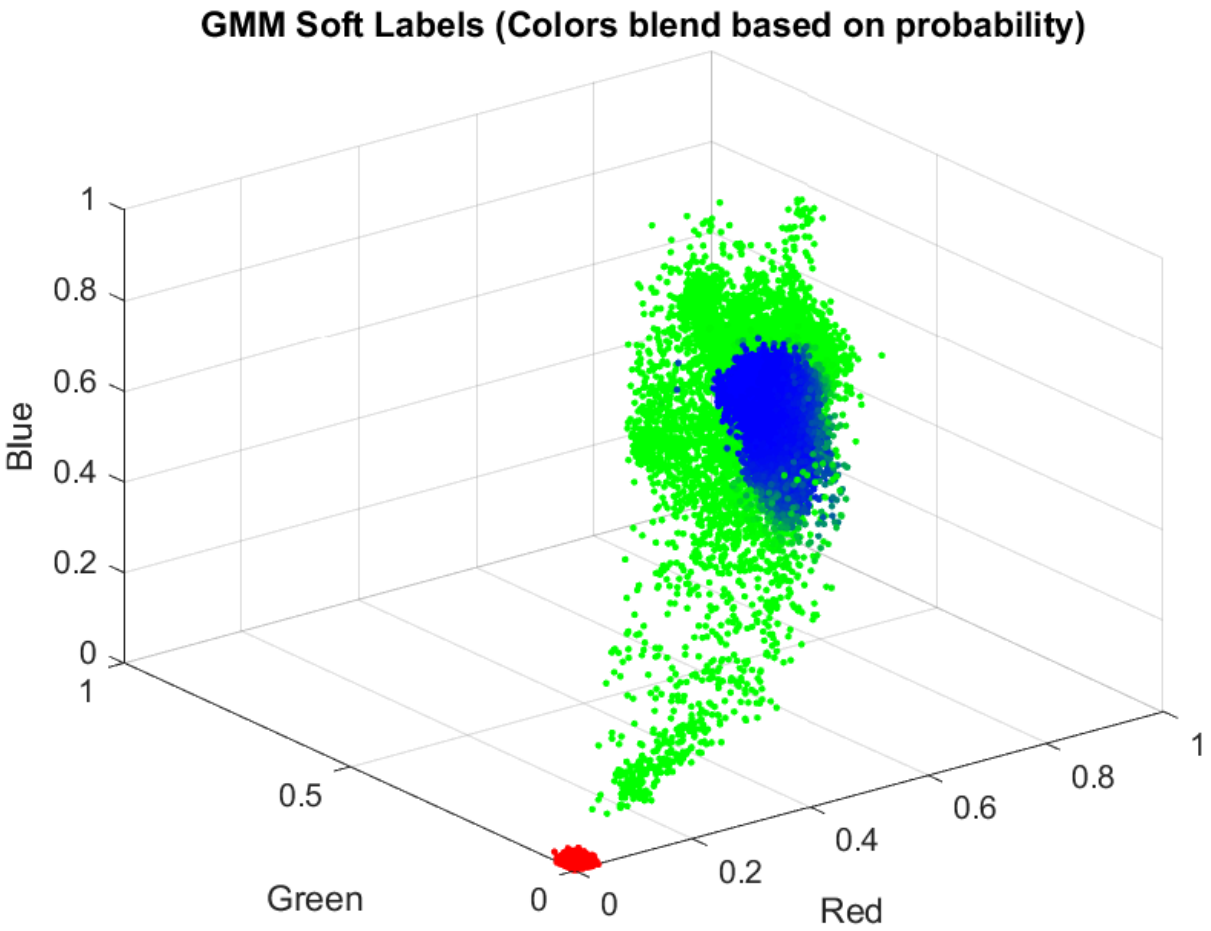
2.4 - Print out the 'fitgmdist' code line, and explain each parameter

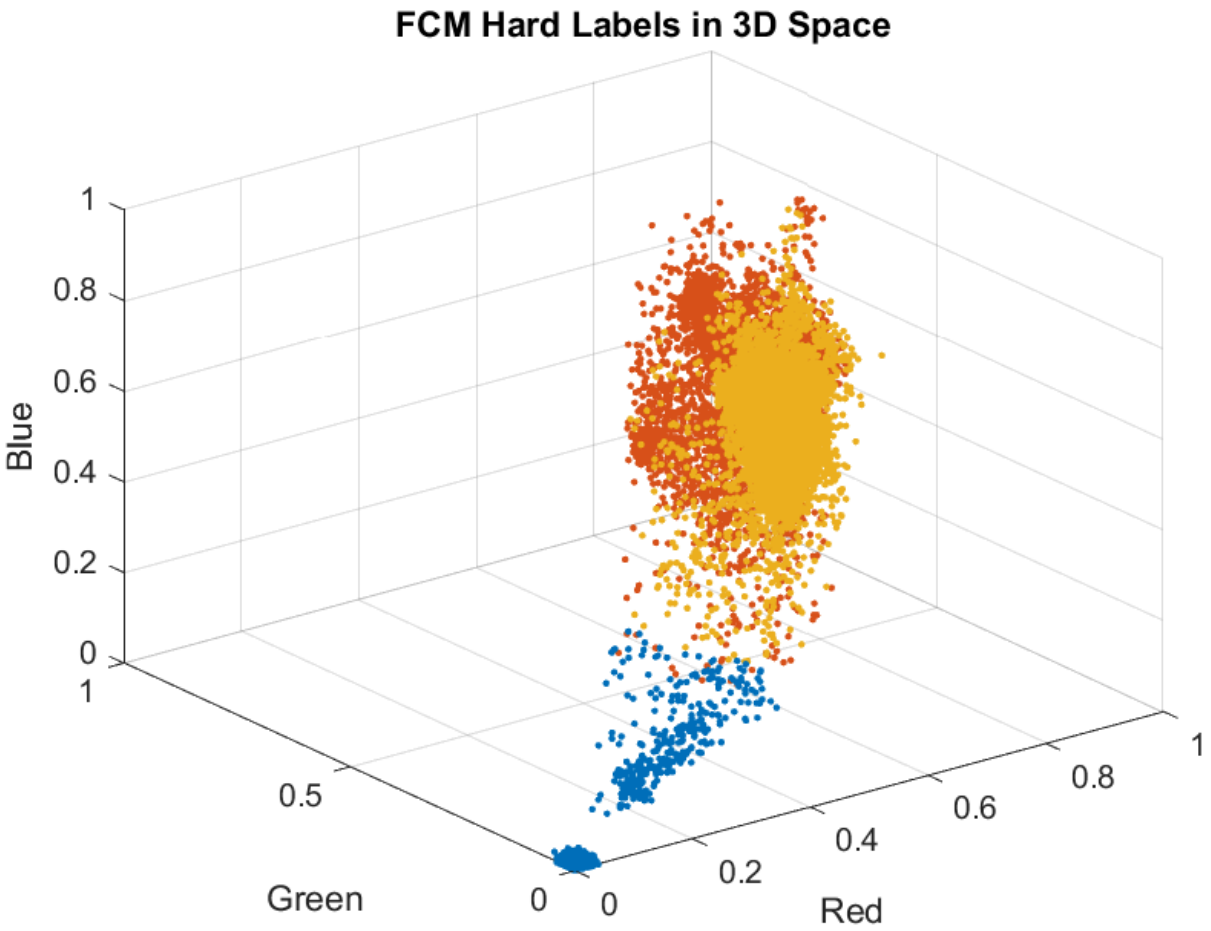
```
gmm_model = fitgmdist(X_noisy, nClusters, 'Options', options_gmm,
'RegularizationValue', 1e-5);
```

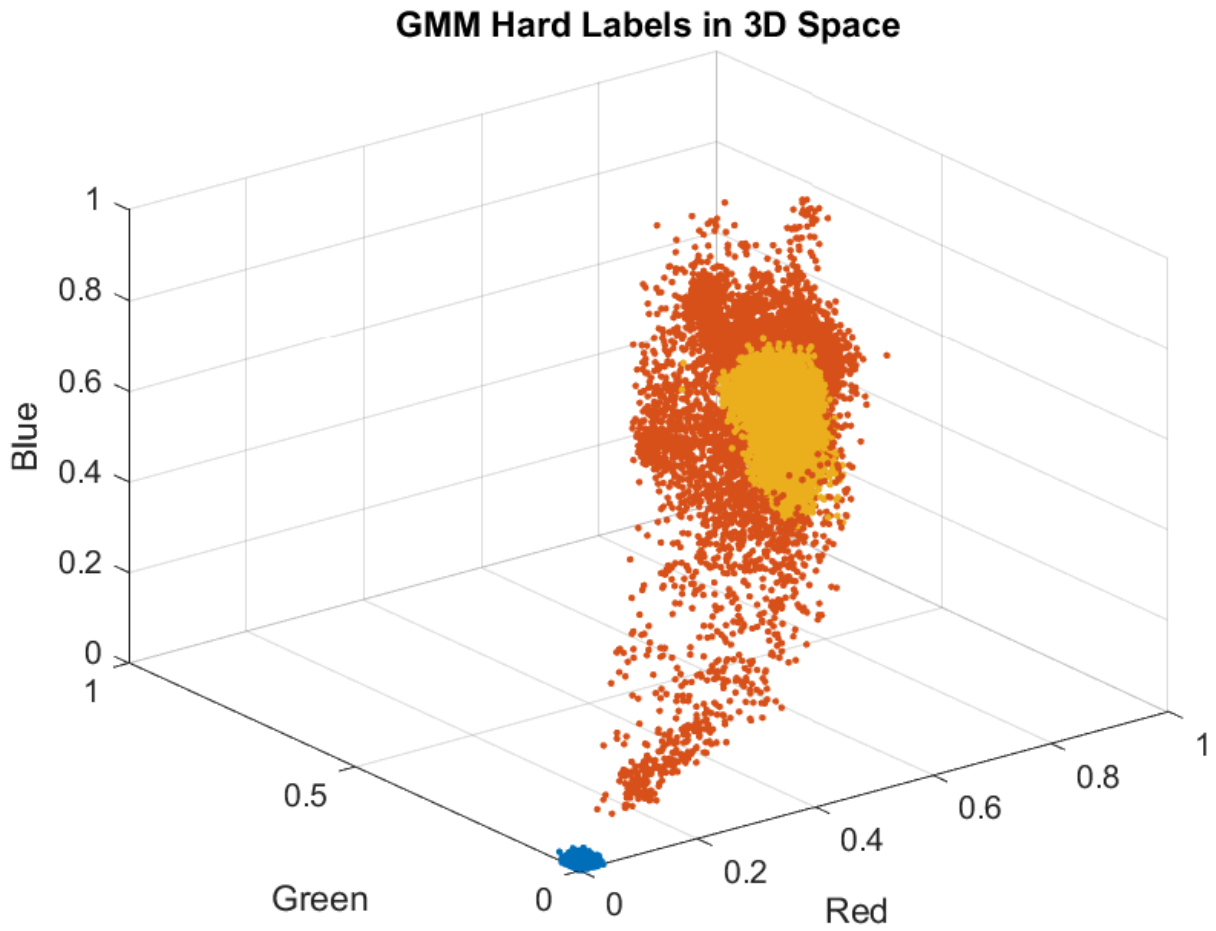
fitgmdist(...): This function fits a Gaussian Mixture Model (GMM) to your data. Bsically it groups your data into clusters using probability.
X_noisy: This is your input data (in your case, the noisy image values reshaped into a dataset).
nClusters: The number of clusters you want the model to find (like how many regions you want to segment the image into).
'Options', options_gmm: These are settings for how the algorithm runs (like max iterations, tolerance, etc.). It controls things like speed and convergence.
'RegularizationValue', 1e-5: This adds a small value to avoid numerical issues (like division by zero or singular matrices). It basically makes the model more stable, especially when data points are very similar

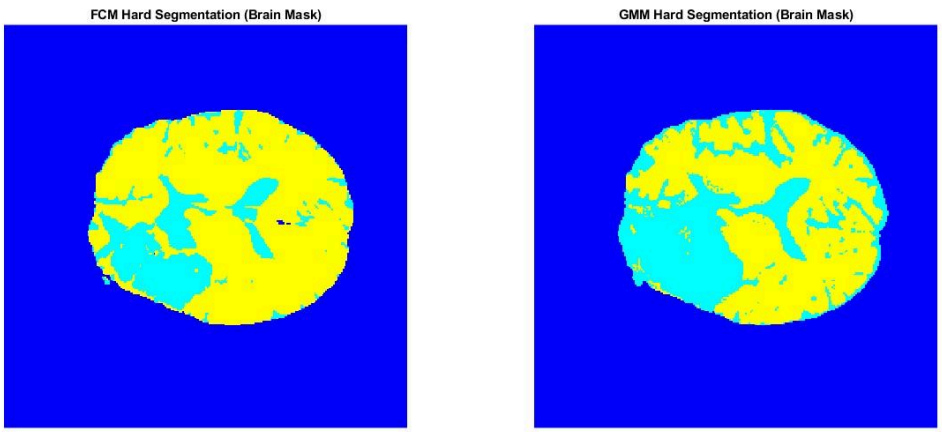

Noisy Image Pixels in 3D Space
Blue Channel
1
0.8
0.6
0.4
0.2
0
Green Channel
1
0.5
0
Red Channel
0
0.2
0.4
0.6
0.8
1


FCM Soft Labels (Colors blend based on membership)
Blue
1
0.8
0.6
0.4
0.2
0
Green
1
0.5
0
Red
0
0.2
0.4
0.6
0.8
1


GMM Soft Labels (Colors blend based on probability)
Blue
1
0.8
0.6
0.4
0.2
0
Green
1
0.5
0
Red
0
0.2
0.4
0.6
0.8
1


FCM Hard Labels in 3D Space
Blue
1
0.8
0.6
0.4
0.2
0
Green
1
0.5
0
Red
0
0.2
0.4
0.6
0.8
1





2.5 - Highlight any ROI between images to describe differences, also describe the difference in methods of converting from soft labels to hard labels.

Looking at regions where colours overlap (ROI), GMM performed better than FCM because it considers the distrib ution (variance) of the data, not just distance. This allowed it to separate clusters more smoothly in areas where values were similar. Both methods start with soft labels, meaning each pixel has a degree of belonging to each cluster. To convert to

hard labels, FCM assigns each pixel to the cluster with the highest membership value, while GMM assigns it based on the highest probability. The key difference is that GMM uses a probabilistic approach, which tends to handle overlap more effectively.

Simulation Exercise 3

3.1 - Provide explanations about the 'Adaptively Regularized Kernel-Based Fuzzy C-Means' method. Briefly summarize the introduction, experiments, conclusion and discussion of this paper (1 paragraph)

This paper proposes an improved version of fuzzy C-means (FCM) called adaptively regularized kernel-based FCM (ARKFCM) to better segment brain MRI images. The main idea is to make clustering more robust to noise by using local image information and a kernel function instead of just basic distance. In the experiments (shown in tables and figures like page 6–9), the method was tested on both simulated and real brain images and consistently performed better than other clustering methods in terms of accuracy and speed. The discussion highlights that the method works well because it adapts to local variations in the image instead of using fixed parameters. Overall, the paper concludes that this approach improves segmentation quality while keeping computational cost low, especially in noisy images

3.2 - Explain each section of the methods of the paper (~3-5 sentences per section – 4 sections)

1. Regularization term (adaptive parameter)

Instead of using a fixed parameter to control how much neighbouring pixels affect clustering, the paper introduces an adaptive parameter based on how different a pixel is from its neighbours. If a region is noisy or varies a lot, the algorithm gives it more attention. This makes the method more flexible since it adjusts automatically depending on the image instead of needing manual tuning. It also helps preserve important details while reducing noise.

2. Weighted image creation

The method creates a new "weighted image" by combining the original pixel values with information from neighbouring pixels. This helps smooth out noise while still keeping important structures visible. The weighting is based on the adaptive parameter, so different parts of the image are treated differently depending on how consistent they are. This step improves segmentation quality without making the algorithm too slow.

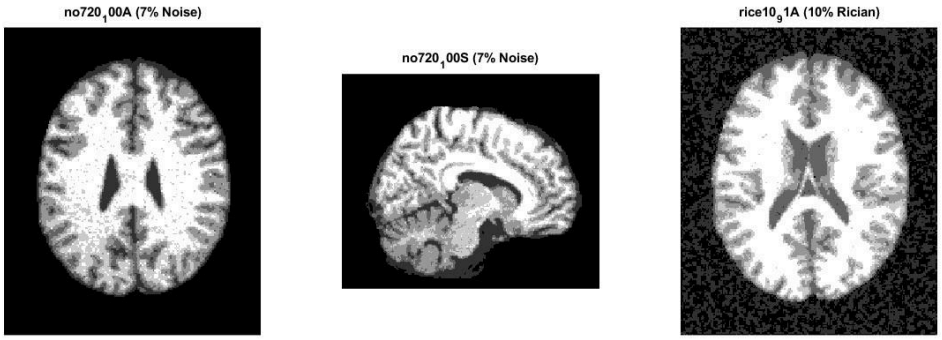
3. Kernel-based distance (instead of Euclidean)

Instead of using normal distance (like Euclidean), the method uses a kernel function to measure similarity. This basically maps the data into a higher-dimensional space where clusters are easier to separate. It makes the algorithm less sensitive to noise and outliers. The

Gaussian kernel is used here because it handles variations smoothly and improves clustering performance.

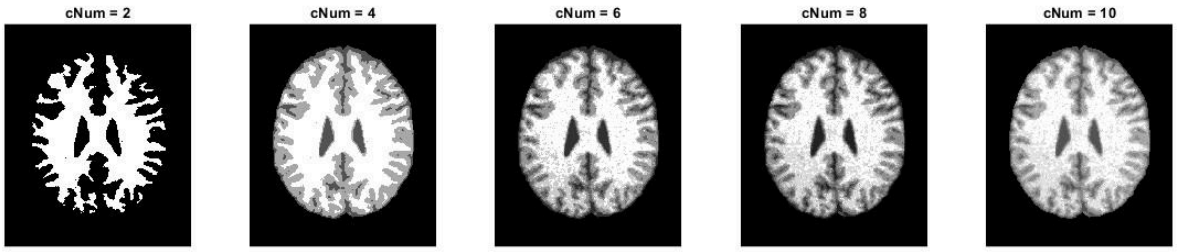
4. Overall ARKFCM framework

The full method combines everything: adaptive regularization, weighted images, and kernel distance into one clustering process. It follows an iterative approach like regular FCM but updates memberships and cluster centres using the improved formulas. The algorithm stops when things stabilize. Variants of the method (like using average, median, or weighted images) give slightly different results, but all aim to improve accuracy while keeping computation efficient.



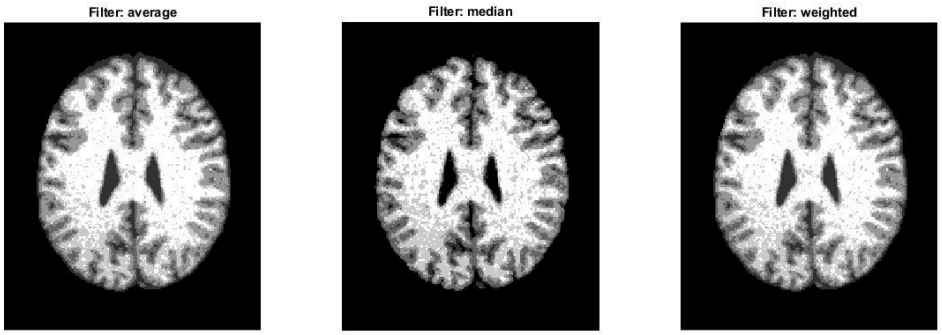
3.3 - Explain how the average filter performs on different noisy images. Is the amount of image noise effective on the performance of the method?

The average filter works pretty well when the noise level is moderate, like in the 7% noise images, because it smooths out small variations while still keeping the main brain structures visible. However, when the noise gets stronger (like the 10% Rician noise image), the filter starts to blur important details along with the noise. This makes edges less sharp and can affect segmentation accuracy. So yes, the amount of noise definitely impacts performance, where higher noise levels reduce how effective the average filter is, since it can't distinguish well between actual structure and noise anymore.



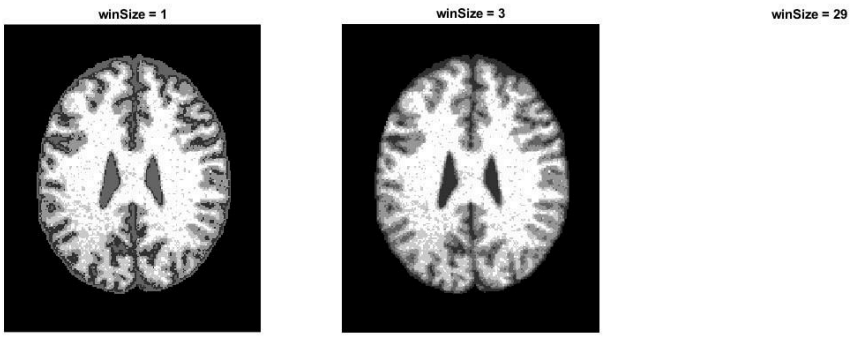
3.4 - Discuss the impact of cNum. Discuss the results and the effect of the number of clusters.

The number of clusters ('cNum') has a big impact on how the image is segmented. When cNum is too low (like 2), the image is oversimplified and different tissues get grouped together, so you lose important detail. As cNum increases (like 4), the segmentation improves because it better separates major regions like white matter, gray matter, and CSF. However, when cNum gets too high (6-10), the image starts to get over-segmented, meaning similar regions are split into multiple clusters and the result becomes noisy or less meaningful. So overall, there's a sweet spot (around 3-4 clusters for this type of image) where you get the best balance between detail and clarity.



3.5 - Discuss the impact of the filters. Which filter is more suitable?

The choice of filter has a clear impact on how well the image is smoothed and how much detail is preserved. The average filter smooths the image but tends to blur edges, which can reduce important structural details. The median filter is better at handling noise (especially speckle or impulse noise) and preserves edges more than the average filter, but it can still introduce some roughness in texture. The weighted filter gives the best balance since it smooths noise while still preserving important features and edges more clearly than the other two. Overall, the weighted filter is the most suitable because it maintains detail while reducing noise effectively.



3.6 Discuss the impact of the window size. What change is created by setting the window size small or large? Why?

The window size controls how much neighbouring information is used when processing each pixel. When the window size is small (like 1), very little neighbourhood information is used, so the image keeps more detail but also keeps more noise. When the window size is moderate (like 3), it provides a good balance, where noise is reduced while important structures are still preserved. When the window size is very large (like 29), the image becomes overly smooth because too many neighbouring pixels are averaged together, which blurs edges and removes fine details. This happens because a larger window mixes more surrounding pixel values, making everything more uniform, while a smaller window focuses more on local detail. So overall, a moderate window size works best since it reduces noise without losing important features.